



POI recommendation by deep neural matrix factorization integrated attention-aware meta-paths

Xiaoyan Li^{1,2} · Shenghua Xu¹ · Hengxu Jin³ · Zhuolu Wang² · Yu Ma² · Xuan He²

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Abstract

With the continuous accumulation of massive amounts of mobile data, point-of-interest (POI) recommendation has become a vital task for location-based social networks. Deep neural networks or matrix factorization (MF) alone are challenging to effectively learn user–POI interaction functions. Moreover, the user–POI interaction matrix is sparse, and the heterogeneous characteristics of auxiliary information are underused. Therefore, we propose an innovative POI recommendation method that integrates attention-aware meta-paths based on deep neural matrix factorization (DNMF-AM). First, we develop a multi-relational heterogeneous information network of “user–POI–geographic region–POI category.” Multiple-weighted isomorphic information networks based on meta-paths are employed to obtain node-embedding vectors across different relationships. Attention networks are employed to aggregate node vectors across various relationships and serve as auxiliary information to mitigate the challenges of data sparsity. Subsequently, the internal embedding vectors of the users and POIs are extracted using feature embedding based on the user–POI interaction matrix. Second, these vectors are integrated with the embedding vectors obtained by aggregating the attention networks. Third, deep neural matrix factorization is used to learn linear and nonlinear user–POI interactions to mitigate the implicit feedback problem. This outcome is achieved using generalized matrix factorization and convolution-constrained multi-head self-attention mechanism deep neural networks. Extensive experiments conducted on two real-world datasets demonstrate that the DNMF-AM outperforms the optimal baseline NeuMF-CAA by 4.24% and 5.04% in terms of HR@10 and NDCG@10, respectively.

Keywords Point-of-interest recommendation · Location-based social networks · Multi-head self-attention mechanism · Heterogeneous information network

Introduction

The growing popularity of location-based social networks (LBSNs), such as Gowalla, Foursquare, Sina Weibo, and Meituan, is attributed to the widespread adoption of GPS-enabled smartphones and the mobile Internet. Point-of-interest (POI) recommendations are essential services in LBSNs owing to their ability to provide intelligent location

services and satisfy personalized user preferences [1]. POI recommendations leverage massive location check-in data in LBSNs to explore aspects of users’ life patterns and preferences concealed by large-scale location data [2, 3]. This allows users to discover interesting personalized locations and generates tremendous commercial benefits for third-party merchants and advertisers [4].

Collaborative filtering (CF) has been employed for personalized POI recommendations in LBSNs. The user–POI interaction matrix is sparse as the user generally visits a small fraction of the vast number of POIs. Mining a user’s behavioral habits using CF is nontrivial [5]. Matrix factorization (MF) [6, 7] is a popular CF method that learns embedding vectors for users and POIs in a common feature space. The dot product is utilized to learn linear user–POI interactions. However, this method is limited to learning low-order relationships and is ineffective for capturing complex higher-order nonlinear relationships.

✉ Shenghua Xu
xushh@casm.ac.cn

¹ Research Centre of Geo-Spatial Big Data Application, Chinese Academy of Surveying and Mapping, Beijing 100830, China

² School of Geomatics, Liaoning Technical University, Fuxin 123000, China

³ Key Laboratory of VGE of Ministry of Education, Nanjing Normal University, Nanjing 210023, China

Recently, deep learning (DL) has been extensively used in POI recommendations owing to its ability to fit arbitrary non-linear functions [8]. Consequently, numerous studies have combined DL with CF to enhance conventional POI recommendations. CF combined with DL primarily learns from two dimensions [9]:

1. The embedding vector learning dimension focuses on modeling users' and POIs' embedding vectors using non-parametric functions to compute their similarity. However, this limits the expressive power of the interaction functions.
2. The interaction function learning dimension models the complex user–POI relationships through neural networks. However, low-order linear interactions are challenging to learn and do not fully leverage the human experience acquired from embedding vectors.

Diverse techniques have been considered to maximize the advantages of different recommendation methods by integrating them for superior results.

Numerous studies have also utilized auxiliary information to alleviate the challenge of data sparsity [10, 11]. The complex and heterogeneous nature of auxiliary information makes it more challenging for POI recommendations to utilize this information [12, 13]. A heterogeneous information network (HIN) can model rich auxiliary information flexibly [14]. The HIN can be used to model auxiliary information for POI recommendations. Relational sequence meta-paths connecting pairs of nodes in a HIN are extensively used to extract the structural features of auxiliary information [15]. By constructing different meta-paths, more information can be effectively integrated to convey different semantics.

Addressing the issues of the complexity of feature interactions being difficult to capture and the heterogeneous relationships of auxiliary information not being fully utilized, we propose an innovative POI recommendation method that integrates attention-aware meta-paths based on deep neural matrix factorization (DNMF-AM). This study utilizes the meta-path-based isomorphic information networks and attention networks to obtain the user and POI multi-semantic auxiliary information representations from the multi-relational HIN of the “user–POI–geographic region–POI category”, which solves the problem of under-utilized heterogeneous relations for auxiliary information. Multichannel learning is performed to capture linear and nonlinear user–POI interactions. Generalized matrix factorization (GMF) is utilized to model user linear preference for POI, whereas convolution-constrained multi-head self-attention mechanism deep neural networks are adopted for modeling user nonlinear preference for POI. This overcomes the difficulty of a single collaborative filter or neural network to effectively capture the complexity of feature interactions

between users and POIs under the influence of implicit feedback.

This study's main contributions are as follows.

- (1) A multi-relational HIN of “user–POI–geographic region–POI category” is developed. Multiple meta-path-based weighted isomorphic information networks are utilized to obtain embedding vectors of nodes in various relationships. These aggregated vectors are ultimately incorporated as auxiliary information by employing attention networks to aggregate node vectors from different relationships.
- (2) The internal embedding vectors of users and POIs are extracted from the user–POI check-in matrix using feature embedding. Subsequently, the internal embedding vectors are integrated with the embedding vectors obtained through attention network aggregation to improve the information richness of the vector representations.
- (3) A novel method for POI recommendation is proposed based on a neural collaborative filtering (NCF) framework. GMF and convolution-constrained multi-head self-attention mechanism deep neural networks are employed to capture linear and nonlinear user–POI interactions. The multi-head self-attention mechanism is adopted for extracting feature correlations, while global self-attention mechanism is restricted by a convolution operation to remove irrelevant terms from the output results.
- (4) Based on the Foursquare dataset (New York City) and the Weibo dataset (Shanghai City), the feasibility and effectiveness of DNMF-AM for POI recommendation are verified.

Related work

POI recommendation is crucial in enhancing the user experience in LBSNs [16]. CF-based POI recommendations are commonly applied to address information overload problems. Furthermore, the popularity of DL in large-scale data and artificial intelligence studies has resulted in the emergence of new hybrid recommendation methods based on DL.

In POI recommendation systems based on CF, Rahmani et al. [17] introduced a spatiotemporal activity-centered model for CF-based POI recommendation that integrates MF to learn user behavior accurately. Subsequently, Rahmani et al. [18] proposed a POI recommendation algorithm that utilizes a localized geographic model and incorporated it into a logistic MF. Cheng et al. [19] combined social information and geographic influence into probabilistic MF. Zhou et al. [20] used temporal information, geographic information, and

social relationships to obtain user preferences for location, and then reconstructed the objective function of the weighted MF by adding implicit feedback terms and regularization terms. However, the user–POI interactions in the aforementioned studies were estimated by calculating the inner product of latent embedding vectors; this did not adequately learn the complexity of user–POI interactions.

DL has gained significant popularity in recommender systems owing to its capacity to address nonlinear relationships, high-dimensional features, and large-scale data [21]. Wang et al. [22] proposed a context- and category-aware POI recommendation model with a dual self-attention mechanism module to learn user preferences for POI and categories in specific contexts. Ding et al. [23] transformed co-visit, geographic, and category data into embedding vectors of POIs and users with feature embeddings. These embeddings were used as pairwise inputs into deep neural networks to adaptively learn high-order nonlinear relationships between features. Zeng et al.'s [24] Neu-PCM, a deep neural network-based POI recommendation model used a union network to extract potential correlations and applied a deep MF to enhance recommendation performance. The NCF proposed by He et al. [25] learned a user–item interaction function using the multilayer perceptron (MLP). Neural MF combined GMF and MLP to address MLP's vulnerability in capturing low-order relationships. Guo et al.'s [26] DeepFM was an end-to-end model that integrates factorization machines (FM) with deep neural networks, modeling low-order interactions of FM into higher-order user–item interactions. Zhang et al. [27] leveraged the self-attention mechanism to deduce relationships between items based on user's historical check-in data, considering global and local intentions inspired by the transformer. Song et al.'s [28] AutoInt used the self-attention mechanism to distinguish between affecting the higher-order features on results, the model could auto-capture higher-order interactions between features and manage high-dimensional sparse data. The aforementioned studies have applied DL networks to enhance the predictive ability of the models by learning user-item interactions. Using neural networks to learn feature interactions directly grants great flexibility to the model. However, without introducing human experience, the learning process is inefficient and prone to local optimization or overfitting.

Therefore, combining the experience of modeling user–POI interactions based on CF with the advantages of DL's complex relationship learning capabilities can significantly improve the performance and generalization of recommender systems. Notably, hybrid approaches of meta-heuristics and machine learning (ML) have yielded significant results in several domains, including hyperparameter optimization [29, 30], feature selection [31], time-series prediction tasks (e.g., prediction of COVID-19 cases [32]), and neural network training [33]. Meta-heuristic algorithms, such

as particle swarm optimization and genetic algorithms [34], can effectively optimize the parameters and structure of ML models by virtue of their global search and adaptive tuning properties, which in turn enhances the predictive accuracy and generalization ability of the models. Meanwhile, these algorithms also play a crucial role in feature selection. More importantly, this approach can incorporate human experience and domain knowledge to provide heuristic guidance for the model. Thus, the hybrid approach of metaheuristics and ML also opens up new possibilities in the field of POI recommendation.

Method

Problem definition

Definition 1 HIN. HIN is defined as a graph $G = (V, E)$, where V denotes the node set and E denotes the edge set. It contains a node-type mapping function $\varphi: V \rightarrow A$ and an edge-type mapping function $\psi: E \rightarrow R$, where $A(|A| > 1)$ is the set of node types and $R(|R| > 1)$ is the set of edge types (Fig. 1a).

Definition 2 Meta-path. A meta-path ξ is denoted by $A_1 \rightarrow^{R_1} A_2 \rightarrow^{R_2} \dots \rightarrow^{R_l} A_{l+1}$, which defines a composite relation $R = R_1 \circ R_2 \circ \dots \circ R_l$ between objects $A_1, A_2, \dots, A_{l+1}$, where $\circ$ is the relation synthesis operator (Fig. 1b).

Definition 3 Weighted neighboring pairs based on meta-path. Provided u_3 and u_1 in a meta-path ξ , if $u_1 \in J_{u_3}^\xi$, then u_3 and u_1 are meta-path-based neighboring pairs $\xi < u_3, u_1 >$ (Fig. 1c). The weighted neighboring pair based on meta-path $\xi < u_3, u_1, w_{u_3, u_1} >$ represents the neighboring pair with weights w_{u_3, u_1} (Fig. 1d).

Definition 4 Weighted isomorphic information network based on meta-path. Considering the meta-path ξ of a HIN G , a weighted isomorphic information network based on meta-path G^ξ is composed of weighted neighboring pairs based on meta-path ξ in the HIN G . Each neighboring pair edge has a corresponding edge weight. Additionally, only one type of node exists in the G^ξ (Fig. 1e).

Definition 5 Check-in matrix. It is formed by the interaction data between users and POIs. The corresponding position r_{up} in the check-in matrix $\mathbb{R}^{M \times N}$ is 1 when user u visits POI p ; otherwise, it is 0, as expressed in Eq. (1). M is the number of users and N is the number of POIs.

$$y_{up} = \begin{cases} 1, & u \text{ interacts with } p \\ 0, & \text{other} \end{cases} \quad (1)$$

Definition 6 POI recommendation. Given a user u and a collection of POI P , POI recommendation is the process of

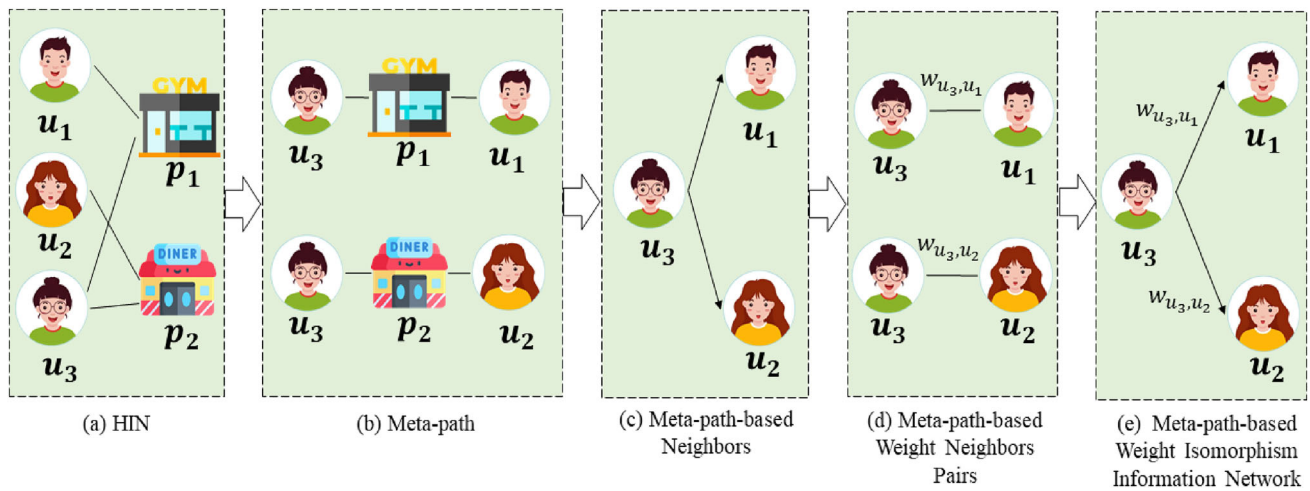


Fig. 1 Conceptual view of defined terms

inferring the first K POIs that user u is most likely to be interested in.

Overall framework

The proposed personalized recommendation method comprised the following steps:

- (1) Constructing a multi-relational HIN of “user–POI—geographic region–POI category;” Utilizing multiple meta-path-based weighted isomorphic information networks to acquire embedding vectors of nodes under different relationships.
- (2) Aggregating node vectors under diverse relationships by utilizing attention networks.
- (3) The internal embedding vectors of users and POIs are extracted from the user–POI interaction matrix through feature embedding. The internal embedding vectors are combined with the aggregated embedding vectors obtained through attention networks.
- (4) Employing GMF and convolution-constrained multi-head self-attention mechanism deep neural networks to capture linear and nonlinear user–POI interactions, respectively.
- (5) Feeding the data from the two aforementioned parts concatenated as a vector into the sigmoid function to generate the predicted results.

Figure 2 illustrates the overall framework.

Developing weighted isomorphic information networks based on meta-paths

This study develops a multi-relational HIN of “user–POI—geographic region–POI category” to extract users and POIs

auxiliary information (Fig. 3). In this study, four types of nodes are selected: (1) user, (2) POI, (3) geographic region, and (4) POI category. Six types of relations are selected: (1) User–POI–User, (2) User–POI–Geographic Region–POI–User, (3) User–POI–POI Category–POI–User, (4) POI–User–POI, (5) POI–Geographic Region–POI, and (6) POI–POI Category–POI.

Transforming meta-paths in a multi-relational HIN into meta-path-based weighted isomorphic information networks could efficiently extract the embedding vectors of the nodes under different relationships. The following section focuses on characterizing the generation process of the user; the generation process of the POI is similar.

Considering a meta-path ξ_m and a user u_i , the set of nodes $J_{u_i}^{\xi_m}$ relevant to the user u_i could be obtained from the meta-path ξ_m . Hypothesis $u_j \in J_{u_i}^{\xi_m}$, u_j could be formed with u_i as a meta-path-directed neighboring pair $\xi_m < u_i, u_j >$. Subsequently, it aggregates all neighboring pairs of all nodes on the meta-path ξ_m to form a meta-path-based isomorphic information network G^{ξ_m} , as expressed in Eq. (2):

$$G^{\xi_m} = g(\{\xi_m < u_s, u_t >, \forall u_s \in V^{\xi_m}, \forall u_t \in J_{u_s}^{\xi_m}\}) \quad (2)$$

where u_s denotes the start node, u_t denotes the end node, V^{ξ_m} denotes the collection of all user nodes on the meta-path ξ_m , $g()$ denotes the integration function. If two neighboring pairs share the same node, that node acts as a connecting node.

In social networks, the affinities between users vary. Consequently, Jaccard similarity [35] is applied to calculate the similarity w_{u_i, u_j} between user u_i and u_j , as presented in Eq. (3):

$$w_{u_i, u_j} = \frac{|J_{u_i}^{\xi_m} \cap J_{u_j}^{\xi_m}|}{(|J_{u_i}^{\xi_m} \cap J_{u_j}^{\xi_m}| + \varpi |I_{i,j}^{\xi_m}|)} \quad (3)$$

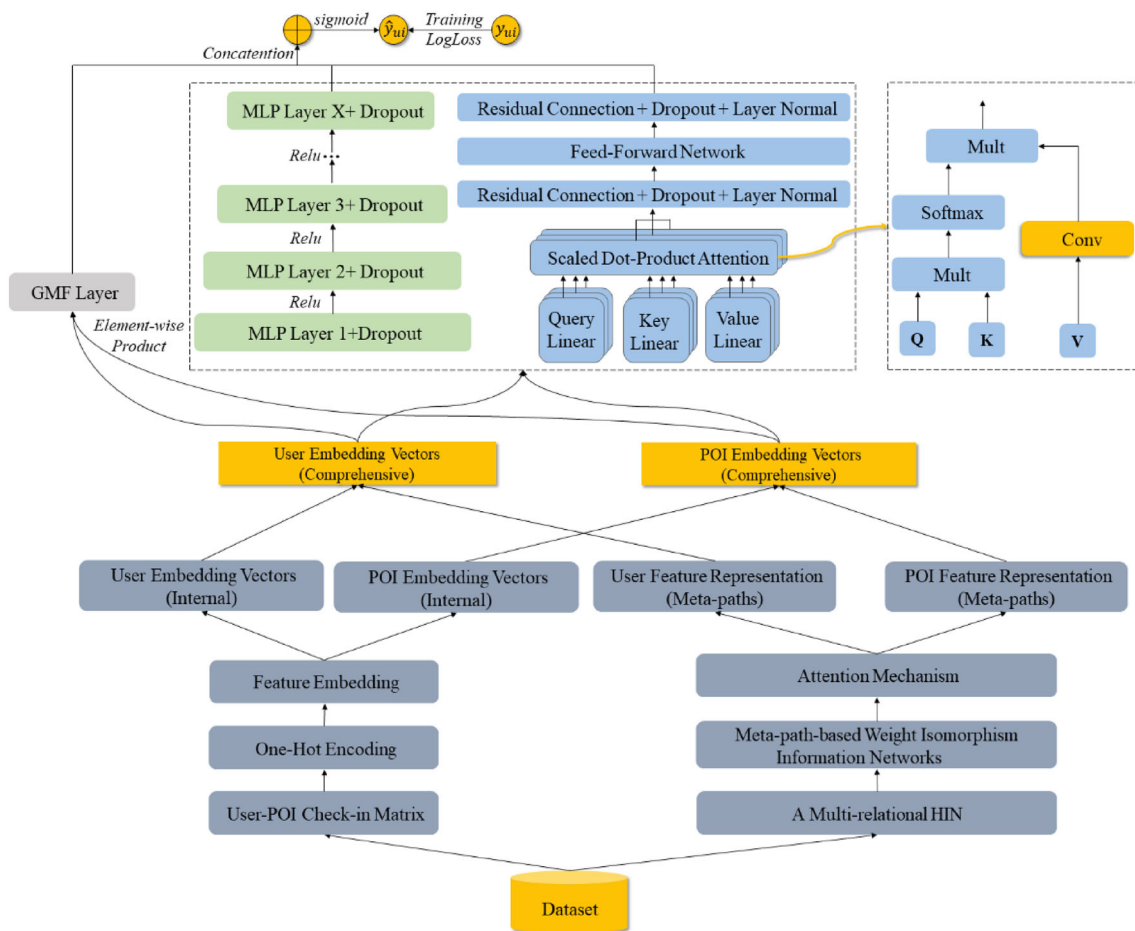


Fig. 2 Overall framework

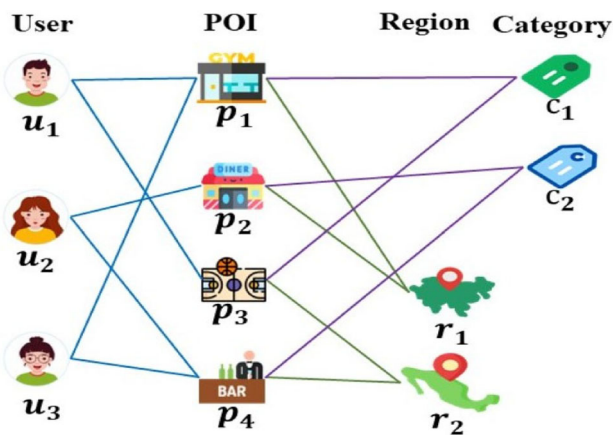


Fig. 3 Multi-relational HIN

where $|I_{i,j}^{\xi_m}|$ denotes the number of paths between u_i and u_j on the meta-path ξ_m , $|J_{u_i}^{\xi_m}|$ and $|J_{u_j}^{\xi_m}|$ denote the number of neighbors on u_i and u_j on the meta-path ξ_m , respectively, w

denotes an adjustable parameter that balance the effect of the number of paths on the similarity.

Thereby, multiple meta-path-based weighted isomorphic information networks $\{G_m^{\xi_m}\}_{m=1}^{|\xi|}$ of the user are obtained, where ξ represents the set of meta-paths in the HIN. Low-dimensional vector spaces are obtained by applying a random-walk algorithm to these networks, resulting in embedding vectors that are both efficient and interpretable. For each edge (u_i, u_j) of $G_m^{\xi_m}$, the joint probability between u_i and u_j is expressed in Eq. (4):

$$P_r(u_i, u_j) = \frac{1}{1 + \exp(-\ell_{u_i}^T \ell_{u_j})} \quad (4)$$

where ℓ_{u_i} and ℓ_{u_j} denote the embedding vector of u_i and u_j , respectively. In $G_m^{\xi_m}$, the empirical joint probability between u_i and u_j is expressed in Eq. (5):

$$\hat{P}_r(u_i, u_j) = \frac{w_{u_i, u_j}}{\sum_{(u_i, u_j) \in E_m^{\xi_m}} w_{u_i, u_j}} \quad (5)$$

where w_{u_i, u_j} is the weight of an edge (u_i, u_j) and E^{ξ_m} is the collection of edges in G^{ξ_m} . To preserve the weights of the edges, we selected the KL scatter of the two probability distributions as the optimization objective, as shown in Eq. (6):

$$O = - \sum_{(u_i, u_j) \in E^{\xi_m}} w_{u_i, u_j} \log p_r(u_i, u_j) \quad (6)$$

By minimizing this optimization objective, the low-dimensional embedding vector $\{h_u^{\xi_m}\}_{u \in U^{\xi_m}}$ of all user nodes on G^{ξ_m} could be obtained, where $h_u^{\xi_m}$ denotes the embedding vector of the node u on meta-path ξ_m .

Attention networks aggregation

Aggregating multiple meta-paths yields the node vectors of users and POIs under different relationships, which can capture their semantic relationships.

Considering a user u and POI p , an embedding vector of u and p on a meta-path can be obtained. $h_u^{\xi_m}$ and $h_p^{\xi_n}$ contain the neighboring and semantic information inside the meta-path, respectively. Consequently, there is a demand to integrate the embedding vectors into all the meta-paths. The effects on the user preferences varies for different semantic networks. Therefore, personalized weights are required to weight such differences. Attention networks could dynamically learn feature weights to achieve the optimal goal. Hence, attention networks are utilized to allocate personalized weights to each semantic network. Equation (7) expresses the attention network implementation.

$$\begin{aligned} a_u^{\xi_m} &= \frac{\exp(\text{relu}(W_{a_u} \times h_u^{\xi_m} + b_{a_u}))}{\sum_{\xi_m \in \xi^{(U)}} \exp(\text{relu}(W_{a_u} \times h_u^{\xi_m} + b_{a_u}))}, \\ a_p^{\xi_n} &= \frac{\exp(\text{relu}(W_{a_p} \times h_p^{\xi_n} + b_{a_p}))}{\sum_{\xi_n \in \xi^{(P)}} \exp(\text{relu}(W_{a_p} \times h_p^{\xi_n} + b_{a_p}))}, \end{aligned} \quad (7)$$

where $a_u^{\xi_m}$ and $a_p^{\xi_n}$ denote the ultimate weights of the meta-paths ξ_m , ξ_n to the user u and POI p , respectively, W_{a_p} and W_{a_u} denote the weights of the attention networks, b_{a_u} and b_{a_p} denote the bias of the networks, $\xi^{(U)}$ and $\xi^{(P)}$ are the collection of meta-paths of u and p , respectively, and relu denotes the activation function.

Based on the attentional weights of each meta-path, the ultimate embedding vectors of the user u and POI p can be

generated, as shown in Eq. (8).

$$\begin{aligned} h_u &= \sum_{\xi_m \in \xi^{(U)}} a_u^{\xi_m} \times h_u^{\xi_m} \\ h_p &= \sum_{\xi_n \in \xi^{(P)}} a_p^{\xi_n} \times h_p^{\xi_n} \end{aligned} \quad (8)$$

Feature embedding and integration

First, ID features are one-hot coded. Second, feature embedding is used to map the initial embedding vectors of the users and POIs into the implicit vector space. Considering this case, the original high-dimensional sparse vectors are transformed into low-dimensional dense vectors, as expressed in Eq. (9).

$$\begin{aligned} H_{U_u} &= \mathcal{h}(U_u) = D^T U_u \\ H_{P_p} &= \mathcal{\lambda}(P_p) = Q^T P_p \end{aligned} \quad (9)$$

where U_u and P_p represent the initial embedding vectors of users and POIs, respectively, $\mathcal{h}$ and $\mathcal{\lambda}$ represent the mapping functions, $D^T \in \mathbb{R}^{M \times \kappa}$ and $Q^T \in \mathbb{R}^{N \times \kappa}$ are the latent factor weight matrices of the users and POIs, respectively, H_{U_u} and H_{P_p} represent the internal embedding vectors of users and POIs, respectively. Subsequently, they are then integrated with the ultimate embedding vectors of the users and POIs obtained through attention networks to acquire comprehensive user and POI embedding vectors, as shown in Eq. (10).

$$\begin{aligned} H^U &= \begin{bmatrix} H_{U_u} \\ h_u \end{bmatrix} \\ H^P &= \begin{bmatrix} H_{P_p} \\ h_p \end{bmatrix} \end{aligned} \quad (10)$$

DNMF

GMF

Traditional MF maps the embedding vectors of users and POIs to the same potential space, evaluating interactions as the inner products of the embedding vectors. In contrast, GMF values their interactions as elemental products of the embedding vectors. The comprehensive embedding vectors of users and POIs are fed into the GMF, and the mapping function of the prediction vector is expressed in Eq. (11):

$$\hat{y}_{Linear} = \phi(H^U, H^P) = H^U \otimes H^P \quad (11)$$

where ϕ is the mapping function of Hadamard product process of the interaction function and $\otimes$ is the element-by-element multiplication between vectors.

Convolution-constrained multi-head self-attention mechanism deep neural networks

User–POI interactions have become increasingly complex owing to the impact of implicit feedback. Therefore, extending the user–POI interactions to nonlinear spaces is crucial.

Left nonlinear layer: MLP exhibits robust learning capabilities that extend user–POI interactions to more complex nonlinear spaces. Thus, this method facilitates user–POI interaction learning. The middle part of the MLP includes several hidden layers. The hidden layer design pattern adheres to the principle of progressively halving the dimensionality. Additionally, to mitigate overfitting and increase the model's generalization capability, we incorporate a dropout layer behind each hidden layer. Equation (12) defines the MLP.

$$Z = \begin{bmatrix} H^U \\ H^P \end{bmatrix} \quad (12)$$

$$y_{MLP} = \alpha_X(\text{Dropout}(W_X^T(\alpha_{X-1}(\text{Dropout}(\dots\alpha_2(\text{Dropout}(W_2^T Z + b_2))\dots)) + b_X))$$

where Z represents the input vector of left nonlinear layer, W_X represents the weight matrix, b_X represents the bias vector, and α_X represents activation function of the layer perceptron. Following the NCF, the *relu* function is adopted as the activation function of MLP, which has non-saturation characteristics and is well adapted to sparse data.

Right nonlinear layer: The multi-head self-attention mechanism can capture nonlinear user–POI relationships, including higher-order interaction patterns and complex dependencies. Different relationships can be addressed using multiple attentional heads. Networks can synthesize different levels of feature interactions, capture richer and more complex interaction patterns between users and POIs, and enhance model interpretability. A local sensory field is introduced in the convolution operation to avoid redundancy of attention units in the global self-attention mechanism. This aspect ensures that each attention unit focuses only on information in the local region. It enables the units to focus on different feature subspaces, enhancing the effectiveness and generalization of the model.

First, the input vector Z is mapped to q , k , and v using linear mapping matrices W^q , W^k , and W^v , respectively, as expressed in Eq. (13).

$$\begin{aligned} q &= ZW^q \\ k &= ZW^k \\ v &= ZW^v \end{aligned} \quad (13)$$

The weights of both q and k^T are calculated using scaled dot-product attention, normalized by the *Soft* max function, and then dot-multiplied by v , which has been convolutionally constrained to generate the attention mechanism output matrix, as expressed in Eq. (14):

$$\text{Attention}(q, k, v) = \text{Soft max}\left(\frac{qk^T}{\sqrt{d_k}}\right)f(v) \quad (14)$$

where $\sqrt{d_k}$ denotes the square root of k . The weights are scaled to avoid excessive magnitudes of molecular dot products. f is convolution operator function, as expressed in Eq. (15):

$$f(v) = \text{relu}(v \odot W_c) \quad (15)$$

where $W_c \in \mathbb{R}^{b \times d_k}$ denotes a convolutional kernel (b denotes the number of convolutional kernels). Then, multiple attention values are computed, and the values are concatenated and multiplied by the weight matrix W_Y to obtain the ultimate matrix Y , as expressed in Eq. (16):

$$\begin{aligned} \text{head}_i &= \text{Attention}(q, k, v) \\ Y &= \text{MHA}(Z) = \text{concat}(\text{head}_1, \text{head}_2, \dots, \text{head}_h)W_Y \end{aligned} \quad (16)$$

where head_i denotes a weight vector denoting the output of the i th attention head, and h is the number of heads of the multi-head self-attention mechanism. Subsequently, a feed-forward network (FFN) with shared parameters is utilized to nonlinearly transform the output information of the multi-head self-attention layer, as expressed in Eq. (17):

$$\text{FFN}(Y) = \text{relu}(YW_1 + b_1)W_2 + b_2 \quad (17)$$

where, W_1 and W_2 denote the linear transformation matrices, and b_1 and b_2 denote bias. Kang [36] cautions that stacking the multi-head self-attention mechanism and FFN might result in overfitting, an unstable training process, and increased training time. Thus, layer normalization, residual connection, and dropout are added after the multi-head self-attention mechanism and FFN, respectively, to solve these problems, as expressed in Eq. (18).

$$\begin{aligned} y_{MHA} &= \text{LN}(\text{Dropout}(\text{MHA}(Z)) + Z) \\ y_{FFN} &= \text{LN}(\text{Dropout}(\text{FFN}(Y)) + Y) \end{aligned} \quad (18)$$

where y_{MHA} denotes the result of the multilayer self-attention layer processed using dropout, residual connection, and normalization. Similarly, y_{FFN} is the result of the processing and the ultimate output.

Eventually, the left and right nonlinear layers are fused using a vector combination, as expressed in Eq. (19):

$$\hat{y}_{NonLinear} = \begin{bmatrix} y_{MLP} \\ y_{FFN} \end{bmatrix} \quad (19)$$

Prediction layers and model training

To efficiently integrate the GMF and convolution-constrained multi-head self-attention mechanism deep neural networks, make them learn separate embeddings respectively and merge them by concatenating the last hidden layer, as shown in Eq. (20):

$$\hat{y}_{up} = \sigma \left(\eta^T \begin{bmatrix} \hat{y}_{Linear} \\ \hat{y}_{NonLinear} \end{bmatrix} \right) \quad (20)$$

where σ represents the *sigmoid* function, and the relationships between them are balanced by the connection weights η . The loss function for model training adopts the binary cross-entropy loss function that is well-suited for implicit data, as expressed in Eq. (21).

$$L = - \sum_{(u,p) \in \gamma \cup \gamma^-} y_{up} \log \hat{y}_{up} + (1 - y_{up}) \log(1 - \hat{y}_{up}) \quad (21)$$

where γ and γ^- represent the collection of positive samples for training and the collection of negative samples after partial sampling of the overall negative samples, respectively. The optimization was performed using the Adam optimizer.

Experiments

Experimental environment and dataset

The study utilized an experimental environment with an Intel(R) Xeon(R) CPU E5-1620 V4 @3.50 GHz and an NVIDIA GeForce RTX 2080Ti. The DL framework was Keras 2.0, and the development language chosen was Python 3.7.

The experiments were performed using two real-world datasets. The first was the publicly available dataset, Foursquare check-ins, within New York City (April 3, 2012, to February 16, 2013) [37]. The Foursquare dataset contains information such as user ID, POI ID, time, latitude and longitude, and POI category. The total number of check-ins in the Foursquare dataset is 172,838 and the data sparsity is 97.55%. The second was a crawled dataset, Sina Weibo

Table 1 Statistics of two datasets

Datasets	Foursquare	Sina Weibo
#users	1083	11,436
#POIs	10,250	17,565
#check-ins	172,838	731,044
#Category	251	130
#Sparsity	98.44%	99.64%

Table 2 Meta-paths used in the Foursquare and Sina Weibo datasets

Meta-paths	Instructions
UPU	Two users visit the same POI
UPCaPU	Two users visit the same category of POIs
UPRePU	Two users visit POIs in the same region
PUP	A user visits different POIs
PCaP	Different POIs belonging to the same category
PRReP	Different POIs belonging to the same region

check-ins, within Shanghai City (January 1, 2019, to November 10, 2021). The Sina Weibo dataset contains information such as user ID, POI ID, time, latitude and longitude. Due to the Sina Weibo dataset lacking POI category information, Python was employed to invoke the AutoNavi Map API for retrieval. The total number of check-ins in the Sina Weibo dataset is 731,044 and the data sparsity is 99.64%. The geographical location information of POIs significantly impacted the users' decision-making behaviors when visiting them. We utilized the *k*-means algorithm to group POIs based on their latitude and longitude information and obtain suitable regional chunks with an assigned location label for their corresponding region. Based on prior work, we excluded users who visited fewer than five POIs and POIs visited by fewer than five users. Table 1 presents the basic statistical information of the datasets, while Table 2 lists the meta-paths used for the two datasets.

Baselines

The effectiveness of the DNMF-AM model was comprehensively illustrated by comparing it with the following recommendation algorithms:

1. MF [38]: The user–POI interaction matrix is factorized into two low-rank matrices representing the implicit features of users and POIs. These features should be learned, and their similarities be calculated.
2. NCF [25]: It combines the GMF and MLP models to make recommendations.

3. CoupledCF [39]: This study extends non-IID learning by introducing a neural coupling approach for capturing explicit and implicit coupling between users and items simultaneously.
4. NeuMF-CAA [40]: This utilizes algorithms like k-means and TF-IDF to extract users' and POIs' auxiliary attribute information, learning their expression and importance with CNN and attention mechanism. Linear and nonlinear user–POI interactions are learned by the GMF and MLP.
5. Neu-PCM [24]: This is a deep POI recommendation method based on neural networks. Critical information is captured by a local learning and dimensionality reduction network. A joint network is applied to extract the potential correlations and a deep MF is optimized for Neu-PCM.
6. CCDSA [22]: It designs a double self-attention mechanism module to explore and capture users' contextual preferences in two different aspects collaboratively, including the fine-grained preference for POI in check-in behaviors and the coarse-grained preference for category.

Experimental setting

A leave-one-out evaluation was employed to assess the performance of implicit feedback-based recommendation methods for POI recommendations. Similar to literature [41], for each user, their most recently interacted POIs were selected as the test data; the rest interaction data as the training data. Additionally, 99 POIs with which the users had not interacted were randomly selected to form the test set [42]. The DNMF-AM method ranked 100 POIs for each user in the test set. The evaluation metrics included the normalized discounted cumulative gain (NDCG) and hit ratio (HR), calculated as expressed in Eq. (22). Recommendation performance improves with higher metric values.

$$NDCG = \frac{1}{M} \sum_{i=1}^M \frac{1}{\log_2(p_i + 1)} \quad (22)$$

$$HR = \frac{\#hits}{M}$$

where $\#hits$ denotes the number of users who obtained accurate recommended results, M denotes the total number of users, and p_i is the position of the real visit value of the i th user on the recommendation list.

During the model training process, we tested with batch size (64, 128, 256, 512), learning rates (0.0001, 0.0005, 0.001, 0.005), number of negatives (1, 2, 3, 4, 5, 6, 7, 8, 9, 10), predictive factors (8, 16, 32, 64, 128), dropout (0.1, 0.2, 0.3, 0.4, 0.5, 0.6), and multi-head number (2, 3, 4, 5, 6,

7, 8). For the Foursquare dataset, the combination of these hyperparameters shows the best performance in the experiments when the batch size is 256, the learning rates is 0.0005, the number of negatives is 4, the predictive factors is 64, the dropout is 0.5 and the multi-head number is 8. For the Sina Weibo dataset, when the batch size is 256, the learning rates is 0.0005, the number of negatives is 4, the predictive factors is 128, the dropout is 0.5 and the multi-head number is 8, giving the best recommendation effect.

Analysis of experimental results

Performance comparison

Figures 4 and 5 illustrate the experimental results for POI recommendation across both datasets. The validity of the DNMF-AM method is illustrated as it outperforms all baselines (where $K = 2, 4, 6, 8$, and 10). Only the case in which $K = 10$ is discussed below. For the Foursquare dataset, Fig. 4 illustrates that the DNMF-AM method achieved an HR of 0.9095 and an NDCG of 0.7887. Compared with the superior-performing NeuMF-CAA, HR is increased by approximately 7.18% and the NDCG by approximately 6.25%. Compared with other baselines, the DNMF-AM method improved the HR and NDCG by approximately 10.20–14.91% and 9.34–15.27%, respectively. For the Sina Weibo dataset, Fig. 5 illustrates that the HR and NDCG of the DNMF-AM method are 0.9228 and 0.8281, which are approximately 4.24% and 5.04% higher than the well-performing NeuMF-CAA in HR and NDCG, respectively. Compared to other baselines, the HR and NDCG of the DNMF-AM model are approximately 7.02–11.85% and 7.35–22.68% higher, respectively. To verify the significance of the performance differences between the different methods, we performed a Kruskal–Wallis test [43]. For the Foursquare dataset, the test shows significant differences between the methods on both HR and NDCG ($H = 31.92$, $p = 1.68e-05 < 0.001$; $H = 32.53$, $p = 1.29e-05 < 0.001$). For the Sina Weibo dataset, the results are equally significant ($H = 30.67$, $p = 2.93e-05 < 0.001$; $H = 32.14$, $p = 1.53e-05 < 0.001$). These tests further confirm that the DNMF-AM method significantly outperforms other baseline in both datasets.

Traditional MF only learns linear user–POI interactions, demonstrating inferior performance in experimental results. This aspect reflects the advantage of employing neural networks to capture user–POI interaction on recommendation performance. CoupledCF treats auxiliary information equivalently and uses only a nonlinear kernel to learn the interaction function. In contrast, DNMF-AM introduces attention networks to differentiate the weight of auxiliary information and utilizes GMF to learn linear user–POI interactions, which can capture user preferences with greater

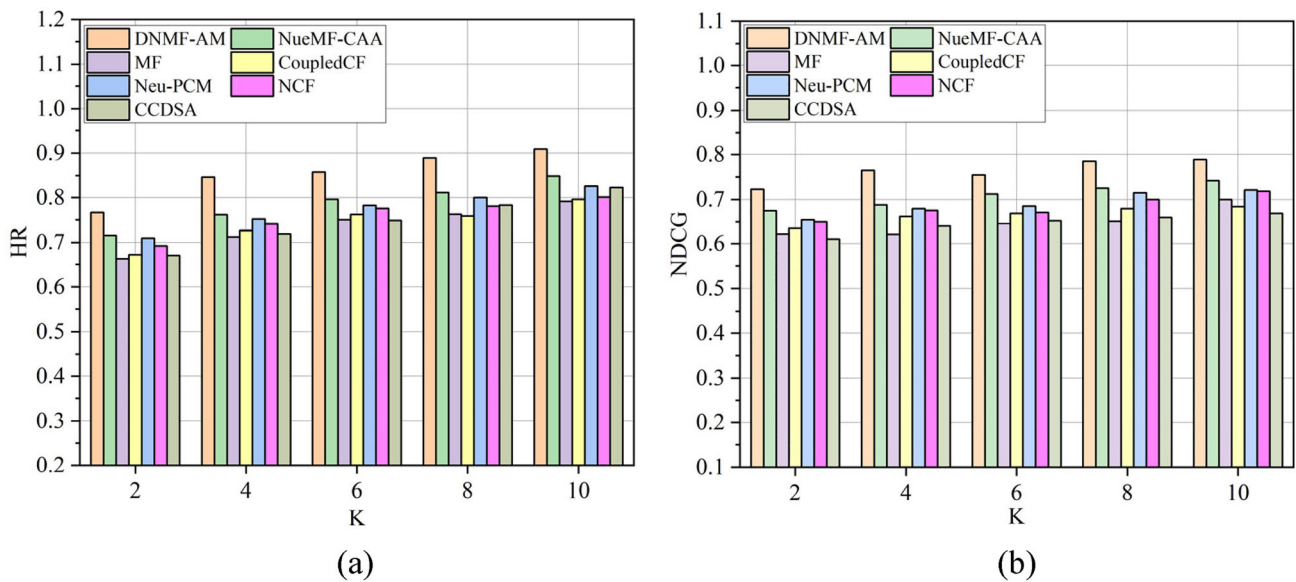


Fig. 4 Performance comparison on Foursquare

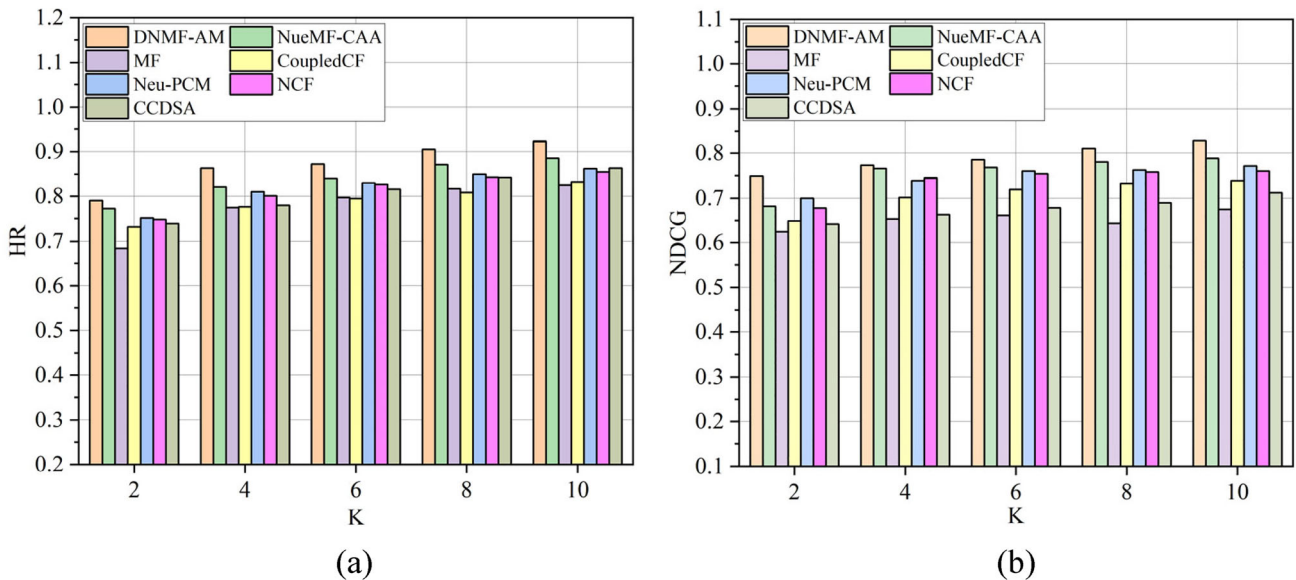


Fig. 5 Performance comparison on Sina Weibo

accuracy. Similarly, DNMF-AM with linear properties outperforms CCDSA, which uses only the dual self-attention mechanism module to learn users' POI preferences and category preferences in different contexts. The NeuMF-CAA generates users' and POIs' auxiliary attribute information with TF-IDF and k-means to acquire richer embedding vectors of users and POIs. NeuMF-CAA outperforms NCF and Neu-PCM, on both datasets, demonstrating that auxiliary information can effectively alleviate data sparsity and improve recommendation performance. DNMF-AM is superior to the better-performing NeuMF-CAA for the following reasons: meta-paths can flexibly obtain abundant auxiliary

information and complex semantic relations from the multi-relational HIN in response to the heterogeneous nature of auxiliary information; DNMF-AM introduces a convolution-constrained multi-head attention mechanism that can distinguish the correlations between different feature domains.

The DNMF-AM performed well on both datasets, demonstrating the excellent generalization capabilities of it. It performs better on the Sina Weibo dataset than the Foursquare dataset owing to the large number of Sina Weibo data and the diversity of user-checked-in POI, which are slightly more singular than the Foursquare dataset.

Ablation study

Ablation studies were conducted to analyze the effectiveness of the different DNMF-AM modules. For clarity, the DNMF-AM variants with different modules were defined as follows:

- (1) -AM: Extraction of auxiliary information from a multi-relational HIN is not considered.
- (2) -GMF: Modeling user–POI interactions using GMF is excluded.
- (3) -ConvMHADNN: Modeling user–POI interactions employing a convolution-constrained multi-head self-attention mechanism deep neural network is excluded.
- (4) -ConvMHA: Modeling user–POI interactions by leveraging a convolution-constrained multi-head self-attention mechanism is excluded.
- (5) -Conv: Convolutional operation is not included.

Figures 6 and 7 illustrate comparative results for the different variants of the DNMF-AM. The results demonstrate that utilizing attention-aware meta-paths to extract the user and POI auxiliary information from the multi-relational HIN significantly improves performance. In the Foursquare dataset, the HR of DNMF-AM is increased by approximately 8.80–10.90%, and the NDCG is increased by approximately 6.96–20.57% compared to -AM. For the Sina Weibo dataset, HR and NDCG are improved by approximately 8.42–10.02%, and 8.17–14.33%, respectively. These results indicate that the auxiliary information generated in this study enables the users' and POIs' embedding vectors to be more accurate and effectively alleviates the data sparsity problem.

Compared with -GMF and -ConvMHADNN, DNMF-AM, which integrates linearity and nonlinearity, exhibits superior expressiveness power on datasets. This outcome further substantiates the effectiveness of integrating linear and nonlinear modeling to capture user preferences. In the Foursquare dataset, the HR of DNMF-AM is increased by approximately 5.93–19.43%, and that of NDCG is increased by approximately 3.68–11.01% compared to -GMF and -ConvMHADNN. In the Sina Weibo dataset, HR and NDCG are enhanced by about 1.30–14.74% and 2.52–27.75%, respectively.

Figures 6 and 7 demonstrate that DNMF-AM, which integrates the convolution-constrained multi-head self-attention mechanism outperforms -ConvMHA. The convolution-constrained multi-head self-attention mechanism module enables multiple linear variations in the input feature representation matrix. Learning the attentional representation of data under different linear transformations enables a more comprehensive internal correlation to be obtained. The evaluation metrics on both datasets are improved by adding a convolution constraint to the multi-head self-attention

mechanism compared to -Conv. This aspect is because the convolution constraint contributes to the enhanced modeling of local features by limiting the scope of the self-attention mechanism to concentrate on information in the local region, thus enhancing model's ability to perform in the local region.

Sensitivity to hyperparameters

(1) Analysis of negative sample rate

Negative samples are introduced as noise to achieve superior model fit during training. Different negative sample rates are tested on the two datasets (Fig. 8). The experimental performance is suboptimal, with the negative sample rate set to one, two, or three, indicating inadequacy in using only one, two, or three negative samples. The best results for both datasets are achieved when the negative sample rate is set to four. These results demonstrate the importance of selecting an appropriate number of negative samples. Insufficient negative samples hinder optimal performance, while excessive negative samples prolong model training and resulted in poorer performance. Consequently, for both datasets, four negative samples are adopted, aligning with He et al.'s [25] experimental results.

(2) Analysis of predictive factors

In this study, the predictive factors have two parts: the internal embedding vectors and the embedding vectors obtained by attention network aggregation. This section discusses the magnitudes of the predictive factors for the internal embedding vectors. Figure 9 illustrates that the model performance on different datasets gradually improved as the predictive factor dimension increased. For the Foursquare dataset, the model performs optimally with the predictive factors of 64. For the Sina Weibo dataset, the model achieves an optimal result with the predictive factors of 128. Different parameter choices for the model yield different recommendation results. In conclusion, the effectiveness of the DNMF-AM depends on the predictive factor that contains adequate representation.

(3) Analysis of different hidden layers in MLP

In this study, MLP is utilized to extract nonlinear user–POI interactions. Therefore, the impact of MLP with different hidden layers on the DNMF-AM method is further investigated. Table 3 indicates that the overall performance improves with an increase in the number of hidden layers (one to three layers). The bold values represent the optimal values. In both datasets, the DNMF-AM method achieves the optimal results with three hidden layers. Mining nonlinear user–POI interactions with the MLP is efficient. Increasing the number of hidden layers contributed to performance improvement. However, in some cases (e.g., when using an MLP with four layers), this does not result in a significant performance

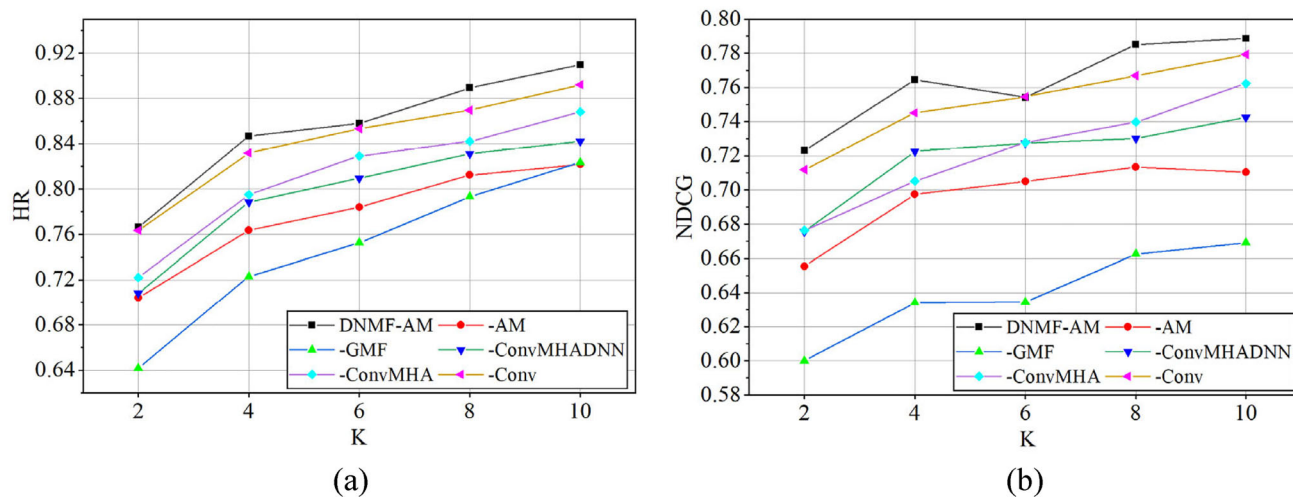


Fig. 6 Ablation experiments on Foursquare

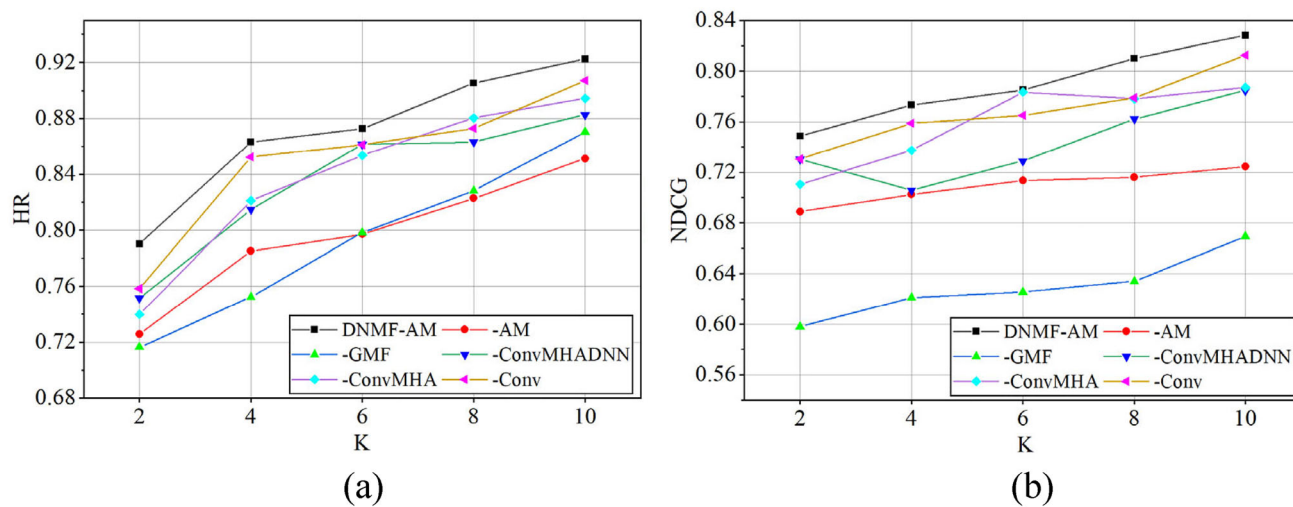


Fig. 7 Ablation experiments on Sina Weibo dataset

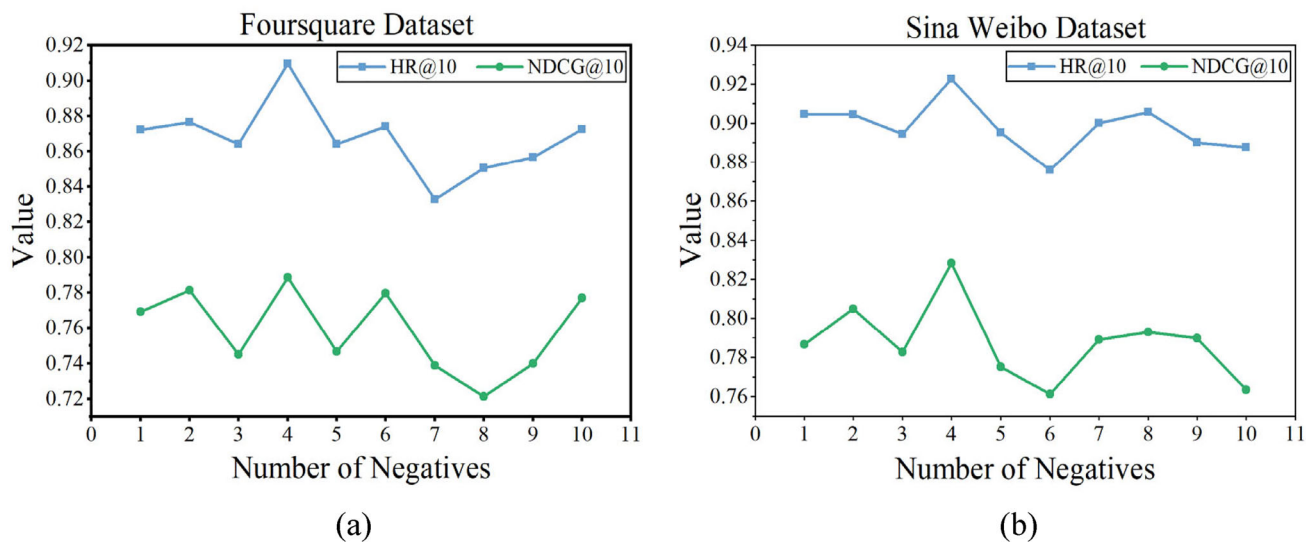


Fig. 8 Experimental results with different negative sampling rate

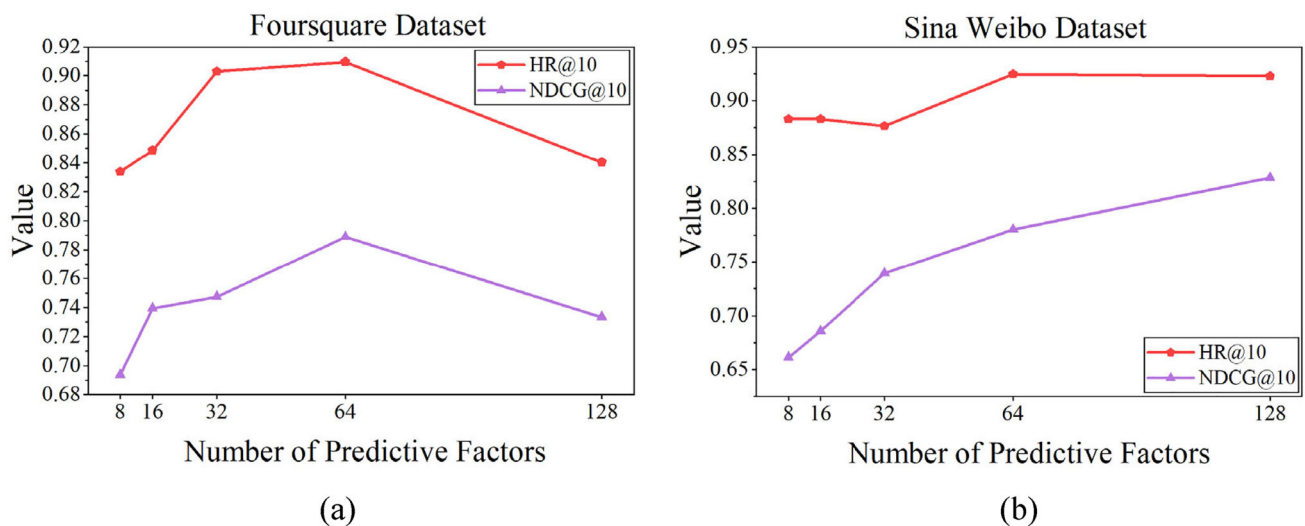


Fig. 9 Experimental results with different predictive factors

Table 3 Experimental results with different hidden layers in MLP

Dataset	Foursquare		Sina Weibo	
	HR @10	NDCG@10	HR @10	NDCG@10
MLP-0	0.8726	0.7524	0.9052	0.7633
MLP-1	0.8846	0.7736	0.8900	0.7752
MLP-2	0.8890	0.7575	0.8948	0.7786
MLP-3	0.9095	0.7887	0.9228	0.8281
MLP-4	0.8800	0.7456	0.9048	0.7936

improvement. This suggests that adding excessive hidden layers is not always beneficial, as it might introduce unnecessary complexity, resulting in overfitting or difficulty training the model.

Conclusions

This study utilizes multiple meta-path-based weighted isomorphic information networks to obtain embedding vectors of nodes in multi-relational HIN of “user–POI–geographic area–POI category” under different relationships. Node vectors from multiple weighted isomorphic information networks are aggregated using attention networks as auxiliary information. Multichannel learning is performed to capture linear and nonlinear user–POI interactions. GMF is used to capture linear user–POI interactions. Convolution-constrained multi-head self-attention mechanism deep neural networks are used to capture nonlinear user–POI interactions. Among them, the multi-head self-attention mechanism differentiates correlations between diverse feature domains, and the introduced convolution operation prevents excessive redundancy of attention units to some extent. Feeding the data

from the two aforementioned parts concatenated as a vector into the sigmoid function to generate the predicted results. Experimental results on the two datasets demonstrate that:

- (1) DNMF-AM outperforms common recommendation methods;
- (2) Attention aware meta-paths could mine users’ and POIs’ auxiliary information under the influence of POI category and geographic information, alleviating data sparsity and using the heterogeneous characteristics of data that are challenging to use;
- (3) Learning the user–POI interactions using the linear properties of GMF and the nonlinear properties of convolution-constrained multi-head self-attention mechanism deep neural networks can effectively alleviate the implicit feedback and enhance the performance of the recommendation algorithm.

DNMF-AM exhibits excellent flexibility and generality, integrates with other linear or nonlinear models, leverages more meta-paths and incorporates various heterogeneous interactions. The performance of POI recommendations is affected by a variety of factors, while DNMF-AM relies

mainly on geographic and POI category factors. Given that user preferences evolve over time, the time factor can be introduced in the future. Additionally, unstructured data such as user review content and check-in images contain abundant information about user behavior and POI, which can be used to analyze user sentiment orientation. Further research is needed to explore how to effectively utilize and integrate this unstructured information.

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Data availability The data supporting this study's findings are available from the corresponding author upon reasonable request.

Declarations

Conflict of interest The authors declared that they had no conflicts of interest with respect to their authorship or the publication of this article.

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